MIL-F-50548A (AR)
AMENDMENT 1
7 January 1983

MILITARY SPECIFICATION

FUZES, ROCKET, M423 AND M427
LESS BOOSTER AND BOOSTER LEAD

This Amendment forms a part of Military Specification MIL-F-50548A (AR), dated 9 November 1978, and is approved for use by the U.S. Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 3

3.3.1: Delete in its entirety, and substitute the following:

"3.3.1 Arming Time - The rotor assembly in the loaded S&A shall rotate from the fully safe position to the fully armed position in not less than 0.67 sec. and not more than .88 sec. when subjected to arming acceleration of 27.00 plus or minus 0.25 g's. The S&A shall meet this requirement at any temperature between -65°F and +165°F."

3.4.1: Delete in its entirety and substitute the following:

"3.4.1 Arming Time - The fuze assembly shall arm in not less than 0.63 sec. or more than 0.92 sec. when subjected to arming acceleration of 27.00 plus or minus 0.25 g's and the detent shall lock the rotor in the fully armed position, as determined by radiographic inspection. The fuze assembly shall meet this requirement at any temperature between -65°F and +165°F."

3.5.1: Delete it in its entirety and substitute the following:

"3.5.1 Arming Time - The rotor assembly in the loaded S&A shall rotate from the fully safe position to the fully armed position in and not less than 1.13 sec. and not more than 1.37 sec. when subjected to arming acceleration of 40.00 plus or minus 0.25 g's. The S&A shall meet this requirement at any temperature between -65°F and +165°F."

FSC 1340

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3.6.1: Delete in its entirety and substitute the following:

"3.6.1 Arming Time - The fuze assembly shall arm in not less than 1.07 sec. and not more than 1.46 sec. when subjected to arming acceleration of 40.00 plus or minus 0.25 g's and the detent shall lock the rotor in the fully armed position as determined by radiographic inspection. the fuze assembly shall meet this requirement at any temperature between -65°F and +165°F.

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4 3.1: Delete "253" and subsitute "376".

PAGE 7

4.3.1: Add "or 9260776" next to "8887706". Delete "8883693" and substitute "9260804". Add "or 9260774 next to "8883692".

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TABLE 1: Delete "253" and substitute "376"; under Fuze, Rocket, Less Booster and Booster Lead, add;

EXAMINATION OR TEST	NUMBER OF SAMPLE UNITS	REQUIREMENTS PARAGRAPH	PARAGRAPH REFERENCE
Low temperature MIL-STD-810 Procedure 1, Method 502	73	3.4.1	4.5.35
High temperature MIL-STD-810 Procedure 1 Method 501	73	3.4.1	4.5.36

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TABLE 1: Delete "253" and substitute "376"; under Fuze, Rocket, Less Booster and Booster Lead, add;

EXAMINATION OR TEST	NUMBER OF SAMPLE UNITS	REQUIREMENTS PARAGRAPH	PARAGRAPH REFERENCE
Low temperature MIL-STD-810 Procedure 1, Method 502	73	3.6.1	4.5.35
High temperature MIL-STD-810 Procedure 1 Method 501	73	3.6.1	4.5.36

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TABLE 1: Delete "(Dwg. 8883706)" and substitute "(Dwg 8883706 or 9260776)".

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TABLE: Delete "8883693" and substitute "9260804".

Delete "(Dwg. 8883692)" and substitute "(Dwg 8883692 or 9260774)".

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4.4.2.16: For critical 8, delete "a" and substitute "d". For major 102, delete "a" and substitute "d". Add NOTE: "d. Samples taken from 5 foot drop, aircraft vibration, non-environmental (including low temperature and high temperature): Note 5' drop not done on every lot.";

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Add:

CATEGORY	EXAMINATION OR TEST	NUMBER OF SAMPLE UNITS	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE
CRITICAL				
10	Low Temperature MIL-STD-810, Procedure 1, Method 502	73	3.4.1	4.5.35
11	High Temperature MIL-STD-810 Procedure 1, Method 501	73	3.4.1	4.5.36

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4.4.2.17: For Critcal 8, delete "a" and substitute "d". For major 102, delete "a" and substitute "d". Add NOTES: "d. Samples taken from 5 foot drop, Aircraft Vibration, Non-Environmental (including Low Temperature and High Temperature): Note 5' drop not done on every lot.";

Add:

CATEGORY	EXAMINATION OR TEST	NUMBER OF SAMPLE UNITS	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE
CRITICAL				
10	Low Temperature MIL-STD-810, Procedure 1, Method 502	73	3.6.1	4.5.35
11	High Temperature MIL-STD-810 Procedure 1, Method 501	73	3.6.1	4.5.36

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4.4.2.20: In M114, delete "visual" and substitute "gage".

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4.4.2.30: Add "or 9260776" next to "8883706".

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4.4.2.32: Delete "8883693" and substitute "9260804".

PAGE 56

4.4.2.34: Add "or 9260774" next to "8883692".

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4.5.7.6, line 5: Delete "to the outside face of the annular gear" and substitute "to the outside face of the annular gear studs".

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4.5.29: Delete in its entirety and substitute:

"4.5.29 Non-Environmental Test - A sample of two hundred and nineteen (219) fuzes shall be selected and subjected to and pass the non-arm test and acceptance criteria specified in 4.5.31. If the test is successful, seventy-three (73) of the 219 fuzes shall be randomly selected and subjected to and pass the arming test and acceptance criteria specified in 4.5.31. The remaining 146 fuzes shall be equally divided into two groups of 73 each. One group shall be subjected to the Low Temperature Test specified in 4.5.35 and the remaining group shall be subjected to the High Temperature Test specified in 4.5.36."

Add:

"4.5.35 Low Temperature Test - A sample of 73 fuzes shall be selected as specified in 4.5.29 and subjected to low temperature environment in accordance with Low Temperature Procedure I at -65°F, Method 502 of MIL-STD-810 with the exception that the time of exposure to temperature shall be a minimum of 4 hours. The fuzes shall then be subjected to and pass arming time test and acceptance criteria specified in 4.5.31 within 5 minutes of removal from Low Temperature.

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NOTE:

Removing the fuze from the cold temperature environment prior to performing the centrifuge test may cause frost to form on the metal parts. This may have an effect on the test results due to attenuation of the acoustic signal of the arming mechanism, thus making it difficult to distinguish the arm signal from background noise. The contractor shall take appropriate action to alleviate this potential problem. One effective method has been to seal each fuze in a plastic bag prior to temperature conditioning. Bagged fuzes are then subjected to the centrifuge test.

4.5.36 High Temperature Test - A sample of 73 fuzes shall be selected as specified in 4.5.29 and subjected to high temperature environment in accordance with High Temperature Procedure I, Method 501 of MIL-STD-810 with the exception that the time of exposure to temperature shall be a minimum of 4 hours. The fuzes shall then be subjected to and pass arming time test and acceptance criteria specified in 4.5.31 within 5 minutes of removal from High Temperature."

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4.5.32: Delete subparagraph a in its entirety and substitute "a. Times under .63 seconds for the M423 and 1.07 seconds for the M427 are classed as arming time under minimum."; Delete subparagraph b in its entirety and substitute "b. Times .93 seconds through 1.30 seconds for the M423 and 1.47 seconds through 1.52 seconds for the M427 are classed as arming time over maximum."

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4.5.32: Delete subparagraph c in its entirety and substitute "c. Times over 1.30 for the M423 and 1.52 for the M427 or fuzes which do not arm or fuzes which arm but do not detent are classified as failure to arm."; Delete subparagraph e in its entirety and substitute "e. All fuzes which are not classed as a failure shall be radiographically inspected prior to firing (vibrated fuzes) or their return to the lot (non-environmental fuzes including hot and cold fuzes)."

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6.3: Delete "DRDAR-QAR-I" and substitute "DRDAR-QAF-I".

6.5: Delete "DRDAR-QAM-M" and substitute "DRDAR-QAF-T".

Delete "DRDAR-LCF" and substitute "DRDAR-LCN-M".

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